

If list is empty

```
If (head == NULL)
    printf("Empty");
```

If the list has only 1 node

If pos == 1

delete-first-node()

If pos == count()

delete-last-node()

If pos < 1 or pos > count()

```
printf("Cannot delete node of this pos");
```

```
If (head->next == NULL)
{
    If (pos == 1)
        delete-first-node();
    Else
        printf("Cannot delete node of this pos");
}
```

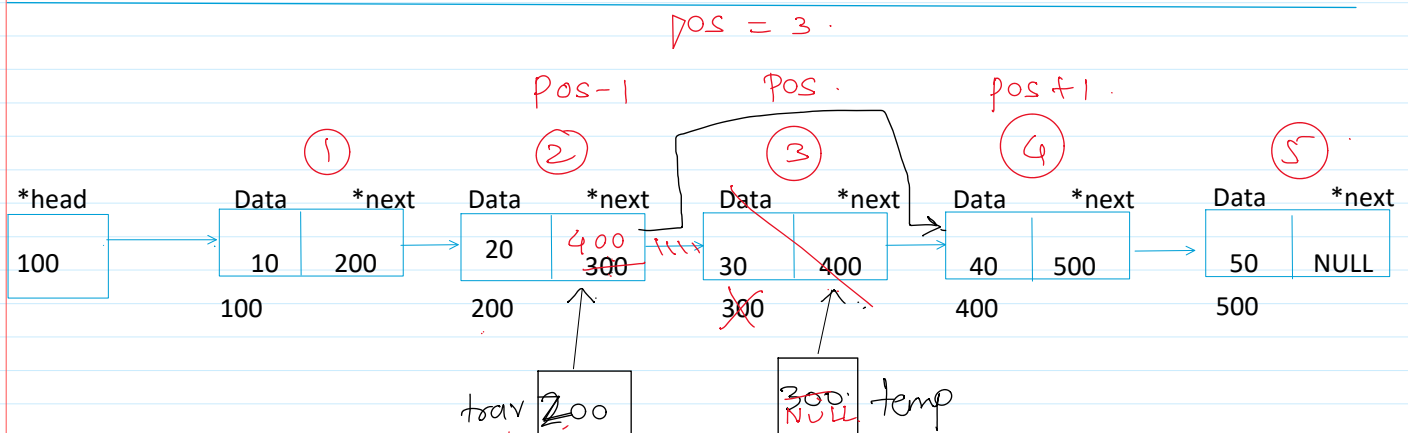
```
If (pos == 1)
    delete-first-node();
```

```
Else
    printf("Cannot delete node of this pos");
```

```
}
```

If pos < 1 or pos > count()

```
printf("Cannot delete node of this pos");
```



1) Take a trav pointer and traverse till pos-1 node.

```
Struct node *trav = head;
```

```
For(int i = 1; i < pos-1; i++)
```

```
Trav = trav->next;
```

2) Take a backup of the pos node in the temp pointer variable.

```
Struct node* temp = trav->next;
```

3) Connect the pos-1 node and pos+1 node.

```
Trav->next = temp->next;
```

4) Free pos node.

```
Free(temp);
```

```
Temp = NULL;
```

```
200->next = 400;
trav->next = 200->next;
temp;
```

```
trav->next = temp->next;
```

Best Case : if pos == 1 : O(1)

Worst Case : pos == count : O(n)

Avg Case : if pos no of the node to be deleted is somewhere in between : O(n)

Add node at first : Best / worst / avg : $O(1)$
Delete first node : Best / Worst / Avg : $O(1)$